

The arc chromatic number of square grids

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Abstract

Let $G_n = [n] \times [n]$, where $[n] = \{1, \dots, n\}$, and let $\chi_{\mathcal{A}}(G_n)$ be the minimum number of colours needed to colour G_n with no monochromatic collinear triple. We prove $\chi_{\mathcal{A}}(G_{11}) = \chi_{\mathcal{A}}(G_{12}) = 7$ and give constructions and proofs for the previously announced values $\chi_{\mathcal{A}}(G_9) = 5$ and $\chi_{\mathcal{A}}(G_{10}) = 6$. We also enumerate optimal colourings up to square symmetry and colour permutation. In particular, $C(6) = 2\,949\,015\,889$, $C(8) = 2$, and $C(9) = 743$. We also show that $\chi_{\mathcal{A}}(G_{8p}) \leq 8p - 8$ for infinitely many primes p .

Keywords: no-three-in-line problem; general position; arc chromatic number; grid colouring; finite fields; computer-assisted proof.

MSC 2020: 05C15, 52C10, 05B25, 68R05.

1 Introduction

A set of planar points is in general position if no three are collinear. The classical no-three-in-line problem [4, 6, 7, 9, 10, 15] asks for the largest such subset of an $n \times n$ integer grid G_n . The trivial upper bound of $2n$ points is attained for every $n \leq 74$ and for $n = 76$ [8, 15], while a heuristic of Guy and Kelly [9] suggests that it is not attained for any large n . Here we study the partition version: the grid is split into general-position sets, and $\chi_{\mathcal{A}}(G_n)$ denotes the minimum number of parts in such a partition. Wood [17] introduced this parameter and proved $\chi_{\mathcal{A}}(G_n) \leq n + p - 1$, where p is the least prime with $p \geq n$, using finite-field parabolas. Araujo-Pardo and Martínez-Sandoval [1, 13] later studied it as the arc chromatic number of the Euclidean grid, determined $\chi_{\mathcal{A}}(G_n)$ for $n \leq 8$, and showed that $\chi_{\mathcal{A}}(G_n) \leq (1 + \epsilon)n$ for every $\epsilon > 0$ and all sufficiently large n . Their first public version left $5 \leq \chi_{\mathcal{A}}(G_9) \leq 6$. The public second version, dated 12 August 2026, records Dong’s subsequent 5-colouring of G_9 , and an independent 5-colouring of G_9 and the announced value $\chi_{\mathcal{A}}(G_{10}) = 6$ of Yoon, Lee, Lim, and Schin [1, Section 6].

The present paper establishes the values for G_{11} and G_{12} , gives explicit witnesses and proofs for G_9 and G_{10} , and classifies the optimal colourings of the small grids. All upper bounds mentioned so far use at least n colours. Section 5 lifts the four-colouring of G_8 through a partition of a finite-field plane and gives $\chi_{\mathcal{A}}(G_{8p}) \leq 8p - 8$ for infinitely many primes p ; in particular $\chi_{\mathcal{A}}(G_n) \leq n - 8$ for infinitely many n , which improves the bound $(1 + \epsilon)n$ of [1] to a bound below n on an infinite set of grid sizes.

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Since a colour occurs at most twice in a row, every colour class has size at most $2n$. Equality in all classes turns the colouring problem into a partition by arcs of size $2n$. This approach was used for G_8 by Araujo-Pardo and Martínez-Sandoval [1, Sections 4.2 and 5]; here it gives further even-grid obstructions and the complete classification of G_8 . We count the G_6 orbits with Burnside's lemma and classify G_9 by exhaustive generation. The search data, representative catalogues and completed computational certificates will be released in the accompanying GitHub repository; see Section 9.

2 Definitions and general bounds

Following Araujo-Pardo and Martínez-Sandoval [1], we write $[n] = \{1, \dots, n\}$ and let G_n be the Euclidean grid with point set $[n] \times [n]$ and collinearity inherited from the Euclidean plane. We identify G_n with its point set when taking subsets or partitions. A *grid line* is the finite set $G_n \cap \ell$, where ℓ is a Euclidean line. When enumerating arc constraints we use the maximal such sets containing at least three points; smaller intersections impose no nontrivial constraint.

Definition 1. An *arc* in G_n is a subset containing no three collinear points. A k -colouring $c : G_n \rightarrow [k]$ is *arc-proper* if each colour class is an arc. The *arc chromatic number* $\chi_{\mathcal{A}}(G_n)$ is the least k for which an arc-proper k -colouring exists. The *profile* of a k -colouring is a list of the sizes of its colour classes, sorted in nonincreasing order.

For a prime power q , we denote the finite field of order q by $\text{GF}(q)$.

Since a row contains at most two points of an arc, every arc in G_n has size at most $2n$. We call an arc of size $2n$ a *full arc*. This differs from a *complete arc*, which usually means an arc that cannot be extended by another point. Hence

$$\chi_{\mathcal{A}}(G_n) \geq \left\lceil \frac{n}{2} \right\rceil. \quad (1)$$

We also use monotonicity,

$$\chi_{\mathcal{A}}(G_n) \leq \chi_{\mathcal{A}}(G_{n+1}),$$

obtained by restricting a colouring of G_{n+1} to an $n \times n$ subgrid. Whenever we wish to check collinearity of points (a_x, a_y) , (b_x, b_y) and (c_x, c_y) we use the determinant criterion

$$\det \begin{bmatrix} b_x - a_x & c_x - a_x \\ b_y - a_y & c_y - a_y \end{bmatrix} = (b_x - a_x)(c_y - a_y) - (b_y - a_y)(c_x - a_x) = 0. \quad (2)$$

Wood [17] proved $\chi_{\mathcal{A}}(G_n) \leq n + p - 1$ when p is the least prime at least n . We record explicitly the prime-modulus bound implicit in the proof of [1, Proposition 5.1]. The construction restricts the standard partition of an affine plane into parabolas [2]; the following direct proof also covers $p = 2$.

Proposition 2 (Parabolic residue colouring). *If $p \geq n$ is prime, then $\chi_{\mathcal{A}}(G_n) \leq p$.*

Proof. Colour $(x, y) \in G_n$ by

$$c(x, y) = y - x^2 \pmod{p},$$

identifying the p residue classes with the colour labels in $[p]$. Suppose that three distinct collinear grid points receive the same residue r . Reduction modulo p preserves both their distinctness, because $p \geq n$, and the vanishing of the determinant (2). Thus the three images are collinear points of $\text{GF}(p)^2$ on the parabola $y = x^2 + r$. A vertical line meets this parabola at most once. A nonvertical line $y = mx + b$ meets it at solutions of the quadratic equation $x^2 - mx + (r - b) = 0$, and therefore in at most two distinct points of $\text{GF}(p)^2$. Hence no line contains three distinct points of the parabola, a contradiction. $\square$

In particular, if $p(n)$ is the least prime at least n , then

$$\left\lceil \frac{n}{2} \right\rceil \leq \chi_{\mathcal{A}}(G_n) \leq p(n).$$

3 Exact values and packing bounds

The known values for $n \leq 8$ from [1, 13] and the values established below are given in Table 1.

n	1	2	3	4	5	6	7	8	9	10	11	12
$\chi_{\mathcal{A}}(G_n)$	1	1	2	2	3	4	4	4	5	6	7	7

Table 1: Exact values of $\chi_{\mathcal{A}}(G_n)$

3.1 Enumeration of optimal colourings for $n \leq 9$

For an optimal colouring of G_n , write $C(n)$ for the number of orbits under $D_4 \times S_{\chi_{\mathcal{A}}(G_n)}$, where D_4 is the order-eight symmetry group of the square and $S_{\chi_{\mathcal{A}}(G_n)}$ relabels the colours. Table 2 uses this convention throughout. We store orbit representatives except for $n = 6$, where the count is obtained directly from Burnside's lemma.

n	$\chi_{\mathcal{A}}(G_n)$	$C(n)$	method
1	1	1	direct enumeration
2	1	1	direct enumeration
3	2	3	direct enumeration
4	2	3	direct enumeration
5	3	62	orbit representatives
6	4	2 949 015 889	Burnside enumeration
7	4	1726	orbit representatives
8	4	2	partitions into full arcs
9	5	743	orbit representatives

Table 2: Counts of the optimal colourings for $n \leq 9$, modulo square symmetries and arbitrary colour relabelling.

For $C(6)$, fix a profile $\lambda = (\lambda_1, \lambda_2, \lambda_3, \lambda_4)$, and for $U \subseteq G_6$ let $L_\lambda(U)$ be the number of ordered decompositions $U = A_1 \dot{\cup} A_2$ into general-position sets with $|A_i| = \lambda_i$. Define $R_\lambda(G_6 \setminus U)$ analogously for the third and fourth classes. The number of colourings with the four profile positions distinguished is therefore

$$\sum_{\substack{U \subseteq G_6 \\ |U| = \lambda_1 + \lambda_2}} L_\lambda(U) R_\lambda(G_6 \setminus U).$$

If m_s is the multiplicity of the class size s in λ , division by $\prod_s m_s!$ removes precisely the remaining permutations of equal-sized colour classes.

For $g \in D_4$, let $F_\lambda(g)$ be the number of such partitions fixed by g , after colours of equal class size have been identified. To make the colour identification explicit, let X_λ be the set of ordered partitions with profile positions distinguished, and put $H_\lambda = \prod_s S_{m_s}$, acting on

equal-sized positions. Then

$$F_\lambda(g) = \frac{1}{|H_\lambda|} \sum_{\sigma \in H_\lambda} |\text{Fix}_{X_\lambda}(g, \sigma)|. \quad (3)$$

Here (g, σ) sends $(A_i)_i$ to $(gA_{\sigma^{-1}(i)})_i$. The H_λ -action is free because the classes are disjoint and nonempty. For each partition fixed after colour identification, every labelling has exactly one compatible σ , proving the formula. Thus symmetries that permute whole equal-sized classes contribute as well as those fixing each class. Writing $A_\lambda = F_\lambda(1)$, Burnside's lemma gives the contribution of the profile as

$$C_\lambda = \frac{1}{8} \left(A_\lambda + \sum_{g \in D_4 \setminus \{1\}} F_\lambda(g) \right).$$

There are 33 possible profiles. The 30 profiles other than the three listed below contribute 933 581 490 orbits. The profiles $(10, 10, 8, 8)$, $(10, 9, 9, 8)$, and $(9, 9, 9, 9)$ contribute respectively 338 037 281, 1 426 881 381, and 250 515 737. Consequently

$$C(6) = 2\,949\,015\,889.$$

The supplement gives the 33 profile contributions and an independent arithmetic audit of the final ledger; it is not a second enumeration. The 9×9 enumeration is described in Section 6.

The contrast between G_6 and G_8 is reflected in the row-capacity slack $2n\chi_{\mathcal{A}}(G_n) - n^2$, which is 5, 12, 7, 0 for $n = 5, 6, 7, 8$ respectively. In particular, every colour class in an optimal colouring of G_8 has size 16, reducing the classification to partitions by full arcs; see Section 4.

3.2 The 9×9 grid

Theorem 3. $\chi_{\mathcal{A}}(G_9) = 5$.

Proof. The row bound gives $\chi_{\mathcal{A}}(G_9) \geq 5$, since four colours cover at most $4 \cdot 18 = 72 < 81$ points. The reverse inequality is witnessed by the explicit matrix in Section 8; its profile is $(17, 17, 16, 16, 15)$. The independent normalized-line checks reported there verify that every colour class is an arc. Thus $\chi_{\mathcal{A}}(G_9) \leq 5$. $\square$

3.3 The even cases $n = 10$ and $n = 12$

Lemma 4. *If n is even and $\chi_{\mathcal{A}}(G_n) = n/2$, then every colour class is a full arc. Thus such a colouring is a partition of G_n into $n/2$ full arcs.*

Proof. Each colour class has size at most $2n$, while $n/2$ classes must cover n^2 points. Equality is forced in every class. $\square$

We use Flammenkamp's frozen complete full-arc catalogues [5]. For $n = 10$, the 156 square-symmetry classes expand to 1135 distinct full 20-point arcs; for $n = 12$, the 566 classes expand to 4348 full 24-point arcs. We independently rechecked the expanded arcs for size and collinearity and reconstructed the stated orbit expansions. Completeness of the input families is that of the cited frozen catalogues.

A hypothetical five-colouring of G_{10} is more constrained. Equality in the capacity bound forces every colour to occur twice in every row and column. Each main diagonal also has ten points, so the same capacity argument forces two points of each colour there. Accordingly, an *admissible full arc* for this obstruction is a full 20-point arc having exactly two points in every row, every column, and each main diagonal. (The row and column conditions already follow

from its size; the diagonal conditions are the additional filter.) Exactly 303 of the 1135 full arcs are admissible in this sense; they form 44 D_4 -orbits.

For G_{10} it is then enough to inspect the 26 admissible full arcs containing the corner $(1, 1)$. For each such arc A , the certificate gives a point $v \notin A$ such that every admissible full arc through v meets A . In a five-colouring, however, the colour class containing v would have to be an admissible full arc disjoint from A . This contradiction rules out all 26 cases. The instances are generated deterministically and include machine-checkable UNSAT proofs.

For G_{12} , the frozen complete catalogue is converted into an exact-cover instance [11] by introducing a binary variable z_A for each of the 4348 full arcs and imposing

$$\sum_{A \ni v} z_A = 1 \quad (v \in G_{12}), \quad (4)$$

together with $\sum_A z_A = 6$. The exhaustive solver reports this 0-1 system infeasible. The supplement records the frozen input, the instance-generation audit, the terminal execution record, and the associated validation checks. Together with the G_{10} obstruction, Lemma 4 gives $\chi_{\mathcal{A}}(G_{10}) \geq 6$ and $\chi_{\mathcal{A}}(G_{12}) \geq 7$.

Theorem 5. $\chi_{\mathcal{A}}(G_{10}) = 6$ and $\chi_{\mathcal{A}}(G_{12}) = 7$.

Proof. The preceding computations give the lower bounds. The stated upper bounds are witnessed by explicit colourings with profiles

$$(20, 20, 20, 20, 10, 10) \quad \text{and} \quad (24, 24, 24, 24, 16, 16, 16)$$

for G_{10} and G_{12} , respectively. The exact matrices are displayed and frozen by digest in Section 8. $\square$

Restricting the G_{12} colouring to any 11×11 subgrid gives $\chi_{\mathcal{A}}(G_{11}) \leq 7$. Together with the row bound, this yields

$$6 \leq \chi_{\mathcal{A}}(G_{11}) \leq 7. \quad (5)$$

3.4 Further packing lower bounds

The catalogue maintained by Flammenkamp [5] gives all full $2n$ -point arcs up to $n = 20$. Its complete $n = 18$ enumeration is due to Chaffin and was independently confirmed by Riley; the $n = 20$ enumeration is due to Prellberg [8]. For $n = 14, 16, 18, 20$, consider the set-packing system

$$\sum_{A \ni v} z_A \leq 1 \quad (v \in G_n), \quad \sum_A z_A = n/2,$$

with $z_A \in \{0, 1\}$. Any feasible solution would select n^2 points in total and hence partition the grid. All four systems are infeasible.

Corollary 6 (Packing lower bounds).

$$\begin{aligned} \chi_{\mathcal{A}}(G_{15}) \geq \chi_{\mathcal{A}}(G_{14}) \geq 8, \quad \chi_{\mathcal{A}}(G_{17}) \geq \chi_{\mathcal{A}}(G_{16}) \geq 9, \\ \chi_{\mathcal{A}}(G_{19}) \geq \chi_{\mathcal{A}}(G_{18}) \geq 10, \quad \chi_{\mathcal{A}}(G_{21}) \geq \chi_{\mathcal{A}}(G_{20}) \geq 11. \end{aligned}$$

Proof. For even n , Lemma 4 shows that a colouring with $n/2$ colours would be a partition of G_n into $n/2$ pairwise disjoint full arcs. The exhaustive set-packing computations rule out such a partition for $n = 14, 16, 18, 20$, giving $\chi_{\mathcal{A}}(G_n) \geq n/2 + 1$ in each case. Since G_n embeds as a subgrid of G_{n+1} , monotonicity gives the adjacent odd-grid bounds. Those odd bounds also follow directly from the row bound $\chi_{\mathcal{A}}(G_{n+1}) \geq \lceil (n+1)/2 \rceil$. $\square$

4 Classification of 4-colourings of G_8

For $n = 8$, equality holds in the row bound, so every colour class is a full arc. It remains only to classify partitions of G_8 into four such arcs.

Theorem 7. *Up to the action of D_4 on the grid and S_4 on the colour labels, there are exactly two arc-proper 4-colourings of G_8 .*

Proof. The frozen complete catalogue has 57 square-symmetry classes of full 16-point arcs, which expand to 380 distinct arcs. We rechecked the expanded arcs for size and collinearity and reconstructed the orbit expansion. Exact-cover enumeration gives four unordered partitions of G_8 into four full arcs; these form two D_4 -orbits. Direct canonicalisation under $D_4 \times S_4$ gives the same two classes, shown in Figure 1. $\square$

2	1	4	3	2	1	4	3
1	3	1	3	2	4	2	4
4	1	2	2	3	3	4	1
3	3	4	1	4	1	2	2
4	4	3	2	3	2	1	1
3	2	1	1	4	4	3	2
2	4	2	4	1	3	1	3
1	2	3	4	1	2	3	4

Previously known orbit

2	1	4	3	4	1	2	3
1	3	1	3	4	2	4	2
4	1	2	4	3	3	2	1
3	3	2	1	2	1	4	4
2	2	3	4	3	4	1	1
3	4	1	1	2	4	3	2
4	2	4	2	1	3	1	3
1	4	3	2	1	2	3	4

Colour-cycling quarter-turn orbit

Figure 1: The two equivalence classes of 4-colourings of G_8 . In the right-hand colouring, a quarter-turn of the grid is compensated by the cyclic permutation $1 \mapsto 2 \mapsto 3 \mapsto 4 \mapsto 1$, after choosing the displayed orientation and labels.

The first orbit is equivalent to the representative displayed by Araujo-Pardo and Martínez-Sandoval [1]. The second was also found independently by Marijn Heule (personal communication to Prellberg).

5 An infinite family with $\chi_{\mathcal{A}}(G_n) \leq n - 8$

The upper bounds of Section 2 use at least n colours. In this section we show that for infinitely many n the grid G_n has an arc-proper colouring with $n - 8$ colours. The construction lifts the four-colouring of G_8 on the left of Figure 1 through a partition of a punctured finite-field plane into arcs with nonsquare chords, and absorbs the 64 points that reduce to the field origin into existing colour classes.

The construction works with residues modulo a prime p , so in this section it is convenient to take $G_n = \{0, \dots, n - 1\}^2$: every point of G_{sp} is then literally $P + pI$ with $P \in G_p$ its residue and $I \in G_s$. Translation by $(1, 1)$ preserves collinearity, so every statement transfers to $[n] \times [n]$. For $v = (a, b) \in \mathbb{Z}^2$ write $N(v) = a^2 + b^2$ and $\|v\|_\infty = \max(|a|, |b|)$, the *height* of v . A nonzero v is *primitive* if $\gcd(a, b) = 1$. Collinearity in a grid always means collinearity in the Euclidean plane; collinearity over a finite field is used only as a proof device.

Let $p \equiv 3 \pmod{4}$ be a prime and let η_p be the quadratic character of $\text{GF}(p)$, that is the Legendre symbol, with $\eta_p(0) = 0$. We identify $\text{GF}(p)^2$ with $\text{GF}(p^2) = \text{GF}(p)(i)$, where $i^2 = -1$, and write $N(z) = z\bar{z}$, so that $N(x + iy) = x^2 + y^2$. Since -1 is a nonsquare in $\text{GF}(p)$, the norm is anisotropic: $N(z) = 0$ only for $z = 0$. A primitive direction $v \in \mathbb{Z}^2$ has *square norm modulo p* if $\eta_p(N(v)) = 1$; for $\|v\|_\infty < p$ the norm $N(v)$ is nonzero modulo p by anisotropy, so this condition only concerns the sign of the character.

Theorem 8. *Let $p \geq 67$ be a prime with $p \equiv 3 \pmod{4}$ such that*

$$\eta_p(a^2 + b^2) = 1 \quad \text{for every primitive } (a, b) \in \mathbb{Z}^2 \text{ with } \|(a, b)\|_\infty \leq 15. \quad (6)$$

Then $\chi_{\mathcal{A}}(G_{8p}) \leq 8p - 8$.

Corollary 9. *There are infinitely many n with $\chi_{\mathcal{A}}(G_n) \leq n - 8$.*

Theorem 8 is proved in Sections 5.1–5.3; Section 5.4 makes the hypothesis (6) explicit and proves the corollary.

5.1 A partition of the punctured plane

A subset of $\text{GF}(p)^2$ is a *field arc* if it is an arc of the affine plane $\text{AG}(2, p)$, that is, no three of its points are collinear over $\text{GF}(p)$; it has *nonsquare chords* if $\eta_p(N(z - z')) = -1$ for all distinct points z, z' in it.

Lemma 10. *Let $p \equiv 3 \pmod{4}$ be prime. Then $\text{GF}(p)^2 \setminus \{0\}$ is the disjoint union of $2(p - 1)$ field arcs with nonsquare chords, each of size $(p + 1)/2$. Exactly $p - 1$ of them, the half-circles, lie on the circles $N(z) = \nu$ with ν a nonsquare, and every half-circle is invariant under $z \mapsto -z$.*

Proof. The norm-one group $U = \{z \in \text{GF}(p^2)^* : N(z) = 1\}$ is the kernel of the surjective norm map $\text{GF}(p^2)^* \rightarrow \text{GF}(p)^*$, hence cyclic of order $p + 1$. Let U^2 be its subgroup of squares, of index two. For each nonsquare $\nu \in \text{GF}(p)^*$ choose β_ν with $N(\beta_\nu) = \nu$ and take the two cosets of U^2 in $\beta_\nu U$. This gives $p - 1$ sets of size $(p + 1)/2$ partitioning the nonsquare-norm region $\{z : \eta_p(N(z)) = -1\}$. Two distinct points of one coset have ratio w^2 with $w = x + iy \in U$ and $y \neq 0$, and

$$N(1 - w^2) = -(w - w^{-1})^2 = 4y^2$$

is a nonzero square, so their chord has norm $4y^2\nu$, of character $\eta_p(\nu) = -1$. Each coset is invariant under negation because -1 , the unique element of order two in U , lies in U^2 : the order $(p + 1)/2$ of U^2 is even. Each coset is a field arc, since a line $\{z + tD\}$ with $D \neq 0$ meets the circle $N = \nu$ in at most two points: $N(z + tD)$ is a quadratic polynomial in t with leading coefficient $N(D) \neq 0$.

For the square-norm region fix $\alpha \in \text{GF}(p^2)$ with $N(\alpha)$ a nonsquare. For $c \in \text{GF}(p)^*$ consider the line

$$L_c = \{z \in \text{GF}(p^2) : \text{Re}(\alpha z) = c\} = \{(c + iu)/\alpha : u \in \text{GF}(p)\}$$

and the inverted set of its square-norm points,

$$H_c = \{z^{-1} : z \in L_c, \eta_p(N(z)) = 1\}.$$

Since $N((c + iu)/\alpha) = (c^2 + u^2)/N(\alpha)$, the retained points are those with $\eta_p(c^2 + u^2) = -1$. There are exactly $(p + 1)/2$ of them, because no summand of $\sum_{u \in \text{GF}(p)} \eta_p(c^2 + u^2)$ vanishes and the sum equals -1 : the number of pairs (u, v) with $v^2 - u^2 = c^2$ is $p + \sum_u \eta_p(c^2 + u^2)$ by character counting and $p - 1$ by the factorisation $(v - u)(v + u) = c^2$. Distinct $z, z' \in L_c$ have $z - z' = i(u - u')/\alpha$, of nonsquare norm, and the identity

$$N(z^{-1} - z'^{-1}) = \frac{N(z - z')}{N(z)N(z')} \quad (7)$$

shows that H_c has nonsquare chords. Every $P \in H_c$ satisfies $N(P) = c^{-1} \text{Re}(P\bar{\alpha})$, that is $N(P - \alpha/(2c)) = N(\alpha)/(4c^2) \neq 0$; so H_c lies on a circle and is a field arc. Finally, the line $\text{Re}(\alpha z) = 0$ consists of the points iu/α , of nonsquare norm for $u \neq 0$. Hence the lines L_c , $c \neq 0$, contain all nonzero points of square norm, and since inversion permutes the square-norm region, the sets H_c partition it. $\square$

The cosets in this proof are the point sets of Kurz [12, Lemma 26] multiplied by an element of nonsquare norm; by [12, Corollary 4] the size $(p+1)/2$ is the maximum for a field arc with nonsquare chords.

5.2 Lifting a seed colouring and absorbing the origin

Let $s \geq 1$, let $A_1, \dots, A_h$ be the colour classes of an arc-proper h -colouring of G_s , the *seed*, and let $p \equiv 3 \pmod{4}$ be prime. Represent the points of $\text{GF}(p)^2$ by their canonical representatives in G_p . Every point of G_{sp} is then uniquely $P + pI$ with $P \in G_p$ and $I \in G_s$; we say that the point *reduces to* P .

Lemma 11 (Lift). *Suppose that every primitive direction of height at most $s-1$ has square norm modulo p . For each part H of Lemma 10 and each $P \in H$ choose a permutation π_P of $[h]$, independently, and colour*

$$P + pI \mapsto (H, \pi_P(j)) \quad (I \in A_j). \quad (8)$$

This is an arc-proper colouring of $G_{sp} \setminus pG_s$ with $2(p-1)h$ colours.

Proof. Let three distinct collinear points have the same colour (H, t) , and consider their residues in H . If the three residues are distinct, they are three collinear points of the field arc H , since the determinant criterion (2) survives reduction modulo p . If all three residues equal P , the three points are $P + pI_1, P + pI_2, P + pI_3$ with I_1, I_2, I_3 collinear in the single seed class A_j , $j = \pi_P^{-1}(t)$. If exactly two residues occur, two of the points are $P + pI_1$ and $P + pI_2$ with $I_1 \neq I_2$, so the primitive direction d of the line satisfies $\|d\|_\infty \leq s-1$ and has square norm modulo p . The third point $P' + pI_3$ has $P' \neq P$, and $P' - P \equiv \lambda d \pmod{p}$ for some integer λ , which is nonzero modulo p ; thus $N(P' - P) = \lambda^2 N(d)$ is a nonzero square, contradicting the nonsquare chords of H . $\square$

The remaining points form pG_s . A fresh copy of the seed palette on them would cost h further colours; the following finite condition on the seed avoids that cost by reusing, for each $T \in G_s$, the tag 1 of a half-circle reserved for T .

Proposition 12 (Absorption). *Let $p \equiv 3 \pmod{4}$ be prime with $p-1 \geq s^2$, and suppose that every primitive direction of height at most $2s-1$ has square norm modulo p . Suppose moreover that for every $T \in G_s$ there are seed colours $j_1(T), j_2(T) \in [h]$, not necessarily distinct, with*

$$A_{j_1(T)} \cap (2T - (1, 1) - A_{j_2(T)}) = \emptyset. \quad (9)$$

Then $\chi_{\mathcal{A}}(G_{sp}) \leq 2(p-1)h$.

Proof. Assign to each $T \in G_s$ a different half-circle $H(T)$ of Lemma 10; there are $p-1 \geq s^2$ of them. Give pT the colour $(H(T), 1)$.

Each $H(T)$ is a disjoint union of antipodal pairs $\{P, -P\}$. Orient every pair in some fixed way. At its first endpoint P choose π_P with $\pi_P(j_1(T)) = 1$, at its second endpoint $-P$ choose π_{-P} with $\pi_{-P}(j_2(T)) = 1$, and complete both permutations arbitrarily. At all other field points use arbitrary permutations. By Lemma 11 the colouring of $G_{sp} \setminus pG_s$ is arc-proper.

Distinct points of pG_s have different colours, so a monochromatic collinear triple would consist of some pT and two points X, Y of colour $(H(T), 1)$, whose residues lie in $H(T)$.

If X and Y reduce to the same P , the primitive direction d of the line has height at most $s-1$, and $X - pT = \lambda d$ with $\lambda \in \mathbb{Z}$. Reducing modulo p gives $P \equiv \lambda d$ with $\lambda \neq 0$, so $N(P)$ is a nonzero square; but P lies on a circle of nonsquare norm.

If the residues differ, they are two points of the circle $N = \nu$ that are collinear with the origin over $\text{GF}(p)$, so they are antipodal. Neither coordinate of a point of nonsquare norm is

zero, so the canonical representative of $-P$ is $(p, p) - P$. Let P be the first endpoint of its antipodal pair. The point of colour $(H(T), 1)$ over P has seed index $\pi_P^{-1}(1) = j_1(T)$, and the one over $-P$ has seed index $\pi_{-P}^{-1}(1) = j_2(T)$, so we may write

$$X = P + pI, \quad Y = (p, p) - P + pJ, \quad I \in A_{j_1(T)}, \quad J \in A_{j_2(T)},$$

and therefore

$$X + Y - 2pT = pW, \quad W = I + J + (1, 1) - 2T, \quad (10)$$

where every coordinate of W lies in $[3 - 2s, 2s - 1]$. Both $X - pT$ and $Y - pT$ are parallel to the line through X, Y, pT , hence so is their sum pW . If $W \neq 0$, the primitive direction d of that line is therefore $\pm W/g$, where g is the greatest common divisor of the coordinates of W ; it has height at most $2s - 1$ and square norm modulo p ; as before $X - pT = \lambda d$ reduces to $P \neq 0$, forcing $N(P)$ to be a square, a contradiction. If $W = 0$, then $I = 2T - (1, 1) - J$ with $I \in A_{j_1(T)}$ and $J \in A_{j_2(T)}$, contradicting (9). $\square$

5.3 The seed and its absorption table

Take $s = 8$, $h = 4$, and as seed the colouring on the left of Figure 1, where the cell with lower-left corner (x, y) carries the label $\kappa(x, y)$; explicitly, with rows $y = 0, \dots, 7$ from top to bottom and columns $x = 0, \dots, 7$,

$$\kappa = \begin{pmatrix} 1 & 2 & 3 & 4 & 1 & 2 & 3 & 4 \\ 2 & 4 & 2 & 4 & 1 & 3 & 1 & 3 \\ 3 & 2 & 1 & 1 & 4 & 4 & 3 & 2 \\ 4 & 4 & 3 & 2 & 3 & 2 & 1 & 1 \\ 3 & 3 & 4 & 1 & 4 & 1 & 2 & 2 \\ 4 & 1 & 2 & 2 & 3 & 3 & 4 & 1 \\ 1 & 3 & 1 & 3 & 2 & 4 & 2 & 4 \\ 2 & 1 & 4 & 3 & 2 & 1 & 4 & 3 \end{pmatrix}. \quad (11)$$

This colouring is arc-proper by Theorem 7; directly, the $4 \binom{16}{3} = 2240$ determinants (2) are nonzero, and each of the four classes determines exactly $\binom{16}{2} = 120$ distinct lines.

The following table supplies the choices in (9): in the same layout as (11), the entry at position (x, y) lists $j_1(T)$ followed by $j_2(T)$ for $T = (x, y)$.

$$\begin{pmatrix} 11 & 11 & 11 & 11 & 11 & 11 & 11 & 11 \\ 11 & 11 & 11 & 11 & 11 & 11 & 11 & 11 \\ 11 & 11 & 11 & 11 & 14 & 33 & 22 & 11 \\ 11 & 22 & 11 & 22 & 14 & 33 & 22 & 11 \\ 11 & 33 & 12 & 34 & 11 & 12 & 12 & 11 \\ 11 & 11 & 22 & 11 & 14 & 44 & 11 & 11 \\ 11 & 22 & 22 & 11 & 23 & 33 & 11 & 11 \\ 11 & 22 & 22 & 22 & 22 & 11 & 11 & 11 \end{pmatrix}. \quad (12)$$

For every $T = (x, y)$, none of the 16 points (u, v) of $A_{j_1(T)}$ has $(2x - 1 - u, 2y - 1 - v)$ in $A_{j_2(T)}$; this is (9), checked in $64 \cdot 16 = 1024$ memberships, or equivalently $(2x - 1, 2y - 1) \notin A_{j_1(T)} + A_{j_2(T)}$. At the eight sites with $j_1(T) \neq j_2(T)$ no single class satisfies (9) with $j_1(T) = j_2(T)$; the freedom to permute the seed colours differently at the two antipodes is what makes absorption possible there.

Proof of Theorem 8. Apply Proposition 12 with $s = 8$ and $h = 4$. The hypothesis (6) is the square-norm condition for heights up to $2s - 1 = 15$, the condition $p - 1 \geq 64$ holds for $p \geq 67$, and the table (12) supplies (9). The resulting colouring of G_{8p} uses $2(p - 1) \cdot 4 = 8p - 8$ colours. $\square$

5.4 Admissible primes

Lemma 13. *For a prime $p \equiv 3 \pmod{4}$, condition (6) holds if and only if $p \equiv 7 \pmod{8}$ and p is a quadratic residue modulo each of the 29 primes*

$$\begin{aligned} &5, 13, 17, 29, 37, 41, 53, 61, 73, 89, 97, 101, 109, 113, 137, \\ &149, 157, 173, 181, 193, 197, 229, 233, 241, 269, 277, 313, 317, 421. \end{aligned} \quad (13)$$

Proof. The list (13) consists of the primes $q \equiv 1 \pmod{4}$ that are sums of two squares of height at most 15. Every prime factor of a primitive $a^2 + b^2$ is 2 or $\equiv 1 \pmod{4}$, and $a^2 + b^2 \leq 15^2 + 14^2 = 421$ when $\|(a, b)\|_\infty \leq 15$. A prime factor $q \equiv 1 \pmod{4}$ of such a sum is either the sum itself, hence in the list, or at most 210, in which case its two-square representation has height at most $\lfloor \sqrt{210} \rfloor = 14$, so it is again in the list.

If $p \equiv 7 \pmod{8}$ and p is a quadratic residue modulo every listed q , then $\eta_p(2) = 1$, and for each listed q quadratic reciprocity gives $\eta_p(q) = \left(\frac{p}{q}\right) = 1$ because $q \equiv 1 \pmod{4}$. Every primitive $a^2 + b^2$ with height at most 15 is nonzero modulo p by anisotropy and is a product of powers of 2 and of listed primes, so (6) holds. Conversely, (6) applied to $(1, 1)$ gives $\eta_p(2) = 1$, whence $p \equiv 7 \pmod{8}$, and applied to a primitive representation of each listed q gives $\left(\frac{p}{q}\right) = \eta_p(q) = 1$. $\square$

Proof of Corollary 9. Let $M = 8 \prod q$ over the primes in (13). Every prime $p \equiv -1 \pmod{M}$ satisfies $p \equiv 7 \pmod{8}$ and $\left(\frac{p}{q}\right) = \left(\frac{-1}{q}\right) = 1$ for every listed q , hence (6) by Lemma 13, and $p > 67$. By Dirichlet's theorem there are infinitely many such primes, and Theorem 8 applies to each of them. $\square$

Remark 14. Heuristically, a proportion 2^{-31} of all primes satisfies the thirty independent conditions of Lemma 13. A search over the residue classes modulo $8 \cdot 5 \cdot 13 \cdot 17 \cdot 29 \cdot 37$ that satisfy the first six conditions shows that the smallest admissible prime is

$$p = 185\,456\,518\,679,$$

so the first grid in the family has side $n = 8p \approx 1.48 \cdot 10^{12}$.

Remark 15 (Limits of the construction). For a seed with h colours on G_s , Proposition 12 saves $sp - 2(p - 1)h = p(s - 2h) + 2h$ colours. Since $h \geq \lceil s/2 \rceil$ by the row bound (1), this saving is at most s , with equality exactly when $h = s/2$, that is, when the seed is a partition of G_s into $s/2$ full arcs (Lemma 4). For $h \geq (s + 1)/2$ the bound $2(p - 1)h$ exceeds sp as soon as $p > s + 1$, so no other seed saves anything. By Table 1 and Corollary 6, the even $s \leq 20$ attaining the row bound are $s = 2, 4, 8$; for $s = 2$ condition (9) fails at $T = (1, 1)$, so the seed of Figure 1 is the largest currently available, and the saving obtainable from Proposition 12 can never exceed the largest s for which G_s is a union of $s/2$ disjoint full arcs. The parts of Lemma 10 cannot be enlarged either, by [12, Corollary 4]. Whether $n - \chi_{\mathcal{A}}(G_n)$ is unbounded remains open.

6 Complete classification of 9×9 five-colourings

We now classify 5-colourings of G_9 under the same $D_4 \times S_5$ action. Equivalently, we classify unordered partitions of G_9 into five arcs, up to square symmetry.

Theorem 16. *There are exactly 743 pairwise inequivalent arc-proper 5-colourings of G_9 under $D_4 \times S_5$; that is,*

$$C(9) = 743.$$

Proof. We perform two independent computations to establish this result. The first exhaustive enumeration proceeds as follows. Generate all arcs of sizes 16, 17, 18, using one representative from each D_4 -orbit of x -arcs for the first colour. If the two largest class sizes are $x \geq y$, only

$$(18, 18), (18, 17), (18, 16), (17, 17), (17, 16)$$

can occur. Indeed, if $x = 18$, then $y \geq 16$, while if $x \leq 17$, then $x = y = 17$ or $(x, y) = (17, 16)$; every omitted alternative has total capacity less than 81.

Fix one of these pairs. Choose an x -arc and a disjoint y -arc. Each of the three remaining final classes has size between

$$81 - x - 3y \quad \text{and} \quad y.$$

The lower bound follows because the other two residual classes have size at most y . We generate all arcs of the allowed integer sizes in the complement, choose two disjoint ones, and take the fifth class to be the remaining points; the candidate is accepted exactly when this fifth class is an arc. Allowing the forced class to exceed the provisional bound y still gives a valid colouring, also covered by the search with its actual two largest class sizes. The five leading-size searches can therefore overlap; final canonicalisation removes duplicate representatives.

Only necessary pruning is used. On a line of length L , the first two classes must cover at least $L - 6$ points and the first three at least $L - 4$, because the remaining three or two colours, respectively, can contribute at most two points each. Equal-sized remaining classes are ordered by their first uncovered point with respect to a fixed total order on G_9 . For any fixed total order this is a value-precedence symmetry break [16]: every unordered choice of equal-sized classes has a unique ordering by its first newly covered point, so no colouring orbit is deleted. The mathematical coverage argument is independent of the choice of total order; the implementation and its ordering convention are part of the computational supplement.

Canonicalisation under $D_4 \times S_5$ leaves 743 orbits, with profiles as in Table 3. A separately written exhaustive implementation, using independently generated arc families, also produces 743 classes under the same equivalence. Thus the independent enumerations agree on the total. All representatives also pass the independent validity and canonicalisation checks. $\square$

Profile	Number of orbits
(18, 17, 17, 15, 14)	2
(18, 17, 16, 16, 14)	13
(18, 17, 16, 15, 15)	22
(18, 16, 16, 16, 15)	37
(17, 17, 17, 16, 14)	38
(17, 17, 17, 15, 15)	43
(17, 17, 16, 16, 15)	384
(17, 16, 16, 16, 16)	204
Total	743

Table 3: The 743 classes of 5-colourings of G_9 , grouped by profile.

7 The 11×11 grid

Equation (5) shows that $6 \leq \chi_{\mathcal{A}}(G_{11}) \leq 7$. It therefore remains to rule out an arc-proper six-colouring of G_{11} .

Remark 17 (Computational evidence). The computer-assisted exclusion below is supported by completed computational packages. These contain the arc catalogues, exhaustive work-item ledgers, terminal records and validation material for all branches. The mathematical reductions, filtering predicates and symmetry arguments are given here; Section 7.5 describes the corresponding computational checks. Jason Dong is assembling the completed packages for release in the GitHub repository described in Section 9.

7.1 Rigidity and the profile theorem

Suppose, for contradiction, that

$$G_{11} = C_1 \dot{\cup} \cdots \dot{\cup} C_6$$

is an arc-proper six-colouring.

Lemma 18 (Long-line rigidity). *On every row, every column, and each of the two main diagonals of G_{11} , the multiplicities of the six colours are*

$$(2, 2, 2, 2, 2, 1)$$

up to permutation.

Proof. A colour occurs at most twice on any line, since three points of one colour on the same line would violate arc-properness. On a line containing eleven grid points no colour can be absent, since the other five colours could then cover at most $5 \cdot 2 = 10$ points. Thus all six colours occur, each once or twice, and their multiplicities sum to 11. The only possibility is

$$(2, 2, 2, 2, 2, 1).$$

□

In particular, every colour occurs once or twice in each row and in each column, so every colour class has size at most 22. Define the *deficit* of C_i by

$$d_i = 22 - |C_i|.$$

Lemma 19 (Deficit–singleton correspondence). *For each colour class C_i , the deficit d_i is both the number of rows in which C_i occurs once and the number of columns in which C_i occurs once. Moreover,*

$$\sum_{i=1}^6 d_i = 11.$$

Proof. Let r_i be the number of rows containing exactly one point of C_i . By Lemma 18, every other row contains exactly two points of C_i . Hence

$$|C_i| = r_i + 2(11 - r_i) = 22 - r_i,$$

and therefore $r_i = 22 - |C_i| = d_i$. Applying the same argument to columns gives the corresponding column statement.

Finally, since the six colour classes partition the 121 points of G_{11} ,

$$\sum_{i=1}^6 d_i = 6 \cdot 22 - \sum_{i=1}^6 |C_i| = 132 - 121 = 11.$$

□

We order the colour classes by nonincreasing size, equivalently the deficits by nondecreasing value. The possibilities with at most one zero deficit are then determined by the integer partitions of 11.

Theorem 20 (Profile theorem). *Up to permutation of the six colours, every arc-proper six-colouring of G_{11} belongs to exactly one of the following three families:*

- $\mathcal{R}$: at least two colour classes have size 22,
- $\mathcal{Q}$: exactly one colour class has size 22,
- $\mathcal{P}$: no colour class has size 22.

The family $\mathcal{Q}$ consists of the following ten profiles:

	$(d_1, \dots, d_6)$	$(C_1 , \dots, C_6)$
Q_1	$(0, 1, 1, 1, 1, 7)$	$(22, 21, 21, 21, 21, 15)$
Q_2	$(0, 1, 1, 1, 2, 6)$	$(22, 21, 21, 21, 20, 16)$
Q_3	$(0, 1, 1, 1, 3, 5)$	$(22, 21, 21, 21, 19, 17)$
Q_4	$(0, 1, 1, 1, 4, 4)$	$(22, 21, 21, 21, 18, 18)$
Q_5	$(0, 1, 1, 2, 2, 5)$	$(22, 21, 21, 20, 20, 17)$
Q_6	$(0, 1, 1, 2, 3, 4)$	$(22, 21, 21, 20, 19, 18)$
Q_7	$(0, 1, 1, 3, 3, 3)$	$(22, 21, 21, 19, 19, 19)$
Q_8	$(0, 1, 2, 2, 2, 4)$	$(22, 21, 20, 20, 20, 18)$
Q_9	$(0, 1, 2, 2, 3, 3)$	$(22, 21, 20, 20, 19, 19)$
Q_{10}	$(0, 2, 2, 2, 2, 3)$	$(22, 20, 20, 20, 20, 19)$.

The family $\mathcal{P}$ consists of the following seven profiles:

	$(d_1, \dots, d_6)$	$(C_1 , \dots, C_6)$
P_1	$(1, 1, 1, 1, 1, 6)$	$(21, 21, 21, 21, 21, 16)$
P_2	$(1, 1, 1, 1, 2, 5)$	$(21, 21, 21, 21, 20, 17)$
P_3	$(1, 1, 1, 1, 3, 4)$	$(21, 21, 21, 21, 19, 18)$
P_4	$(1, 1, 1, 2, 2, 4)$	$(21, 21, 21, 20, 20, 18)$
P_5	$(1, 1, 1, 2, 3, 3)$	$(21, 21, 21, 20, 19, 19)$
P_6	$(1, 1, 2, 2, 2, 3)$	$(21, 21, 20, 20, 20, 19)$
P_7	$(1, 2, 2, 2, 2, 2)$	$(21, 20, 20, 20, 20, 20)$.

Proof. By Lemma 19, the deficits are nonnegative integers with sum 11. If at least two deficits are zero, the colouring belongs to $\mathcal{R}$. If exactly one deficit is zero, the remaining five deficits are positive integers with sum 11. Their nondecreasing partitions are exactly the ten profiles $Q_1, \dots, Q_{10}$ listed above. If no deficit is zero, all six deficits are positive integers with sum 11. Their nondecreasing partitions are exactly the seven profiles $P_1, \dots, P_7$. These three alternatives are disjoint and exhaustive. $\square$

7.2 Large arcs, symmetry, and profile transfer

For the computational part we work with the large colour classes rather than with cell assignments.

An *anchor* is a fixed first colour class, and an arc prefix is an ordered list of fixed classes. In the implementation, a *mask* is a bit-vector encoding a point subset, and a *shard* is one part of a disjoint division of the search domain. SAT and UNSAT mean satisfiable and unsatisfiable, respectively. A CNF formula is a conjunction of clauses; a not-all-equal (NAE) constraint requires its Boolean variables to take both values. The Davis–Putnam–Logemann–Loveland (DPLL) procedure [3] is used for exact Boolean search.

Large-arc catalogues. Lemma 18 gives necessary conditions on any arc occurring as a colour class in a six-colouring. A size-22 arc contains exactly two points in every row and

every column, and it must meet each of the two main diagonals. There are 1 120 size-22 arcs in G_{11} ; exactly 676 meet both main diagonals. These 676 admissible arcs form 89 orbits under the dihedral group D_4 of the square.

Let $\mathcal{C}_{22}$ denote this family of admissible size-22 arcs. Likewise, let $\mathcal{C}_{21}$ denote the frozen catalogue of *diagonal-feasible* size-21 arcs satisfying the corresponding necessary conditions for a colour class: exactly one singleton row, exactly one singleton column, and nonempty intersection with each main diagonal. Here and below *oriented* means that the coordinates are fixed and the objects have not yet been quotiented by square symmetry. The oriented catalogue contains

$$|\mathcal{C}_{21}| = 1\,019\,640$$

arcs, forming 127 491 orbits under D_4 . Fix sets $\mathcal{T}_{22} \subseteq \mathcal{C}_{22}$ and $\mathcal{T}_{21} \subseteq \mathcal{C}_{21}$, each containing one representative of every D_4 -orbit. Thus $|\mathcal{T}_{22}| = 89$ and $|\mathcal{T}_{21}| = 127\,491$.

The computational packages use the term *cap* for a planar arc; we retain their literal artifact identifiers but use *arc* throughout the mathematical exposition. Any zero-based grid coordinates in those packages correspond to $(x-1, y-1)$ for the manuscript point (x, y) . This translation preserves collinearity and all square-symmetry orbits.

All subsequent searches use these frozen catalogues. Their validity, completeness, and D_4 -orbit decompositions are checked separately; see Section 7.5.

Symmetry of arc prefixes. Square symmetries act simultaneously on all colour classes: a single element of D_4 is applied to every arc in a tuple. We may also permute only those colour classes that have identical prescribed sizes, roles, and additional constraints in the current search problem. In particular, the tuple quotient is not obtained by canonicalising each constituent arc under an independently chosen square symmetry.

Let

$$\mathbf{A} = (A_1, \dots, A_r)$$

be a prefix of pairwise disjoint arcs, and let H be a group generated by such square symmetries and admissible colour permutations.

Lemma 21 (Symmetry reduction). *If two arc prefixes lie in the same H -orbit, then one can be completed to a valid colouring satisfying the prescribed profile and constraints if and only if the other can.*

Proof. Every element of D_4 is a bijection of G_{11} preserving collinearity. An admissible colour permutation merely relabels classes whose prescribed roles and constraints are unchanged. Applying the same grid symmetry and colour permutation to a completion therefore gives a bijection between the completions of the two prefixes. $\square$

We retain one representative from each relevant prefix orbit and call it a *canonical prefix*. Canonicalisation acts on the prefix as a whole; independent canonicalisation of its constituent arcs need not preserve the tuple orbit because stabilisers of earlier arcs may act on later choices.

Corollary 22 (Orbit coverage). *Let X be a finite H -invariant family of arc prefixes. Any subset $Y \subseteq X$ meeting every H -orbit preserves the existence or nonexistence of a completion with the prescribed profile and constraints.*

Proof. By Lemma 21, completability is constant on each orbit. $\square$

The symmetry quotients are $D_4 \times S_3$ for the three equal-role size-21 classes in P_4, P_5 , and $D_4 \times S_2$ for the two such classes in P_6 . The five residual size-20 classes in P_7 are treated as an unordered packing. One may generate the complete prefix universe before quotienting, or use

canonical generation with a proof that every relevant orbit is represented; see [14] for general canonical-generation methods. The particular K_3 and K_2 rules used here and their coverage arguments are given in the branch proofs below.

Transfers between profiles. Previously eliminated profiles can also restrict later ones.

Lemma 23 (Profile transfer). *Let*

$$G_{11} = C_1 \dot{\cup} \dots \dot{\cup} C_6$$

be an arc-proper colouring. Let $A = C_i$ and $B = C_j$ be distinct colour classes, and let $D \subseteq B$. If $A \cup D$ is an arc, then replacing

$$A \text{ by } A \cup D, \quad B \text{ by } B \setminus D$$

and leaving the other classes unchanged gives another arc-proper colouring.

Hence, if no arc-proper six-colouring exists with the resulting multiset of class sizes, then the specified transfer is impossible in the original colouring.

Proof. The modified classes remain pairwise disjoint and still cover G_{11} . The recipient $A \cup D$ is an arc by assumption, while $B \setminus D \subseteq B$, so deleting points from B cannot create a collinear triple. The other four classes are unchanged. $\square$

The donor class and the transferred set are part of the hypothesis. In particular, a transfer exclusion may not be replaced by an unrestricted maximality or saturation condition unless that stronger statement has been proved separately.

As an example, take a size-21 class in profile P_4 . Moving one point into it from a class of size 21, 20, or 18, whenever the enlarged class remains an arc, would give respectively

$$(22, 21, 20, 20, 20, 18) = Q_8,$$

$$(22, 21, 21, 20, 19, 18) = Q_6,$$

$$(22, 21, 21, 20, 20, 17) = Q_5.$$

Thus, after Q_5, Q_6 , and Q_8 have been eliminated, Lemma 23 implies that every size-21 class in P_4 is nonextendible by any point of G_{11} outside that class.

Similarly, for P_5 , one-point transfers from classes of sizes 21, 20, and 19 lead respectively to Q_9, Q_7, Q_6 . These consequences are used only after the corresponding target profiles have been eliminated.

Lemma 24 (Monotone order for one-point transfer dependencies). *For a six-colouring with class sizes $s_1, \dots, s_6$, put*

$$\Phi = \sum_{i=1}^6 s_i^2.$$

Suppose a transfer used as a pruning implication moves one point from a donor class of size $b \leq 21$ into a recipient class of size 21. Then the resulting profile has strictly larger Φ ; indeed,

$$\Delta\Phi = 22^2 + (b-1)^2 - 21^2 - b^2 = 44 - 2b > 0.$$

Consequently any dependency chain consisting solely of such one-point transfers is acyclic.

All one-point profile transfers used below for P_4, P_5 , and P_6 have this form, so their dependency order is covered by Lemma 24. The reductions use only the stated donor-specific moves and their proved consequences.

After a large-arc prefix is fixed, the remaining classes are handled as a residual arc partition.

7.3 Exact residual reductions

Fix pairwise disjoint colour classes $A_1, \dots, A_r$ and put

$$R = G_{11} \setminus (A_1 \cup \dots \cup A_r).$$

If the remaining class sizes are $q_1, \dots, q_m$, then a completion is equivalently a partition

$$R = B_1 \dot{\cup} \dots \dot{\cup} B_m, \quad |B_i| = q_i,$$

in which every B_i is an arc.

For a residual set R , write

$$\mathcal{A}_s(R) = \{B \subseteq R : |B| = s, B \text{ is an arc}\}.$$

Selecting the remaining classes from these complete families, subject to pairwise disjointness and their prescribed sizes, gives an exact formulation of the residual partition problem. A production search may use smaller families when its filters are necessary for a completion.

Definition 25 (Coverage-sound branch family). Fix a computational branch, a retained prefix $\mathbf{A}$, and a required residual class size s . A candidate family $\mathcal{C}_s^{\text{branch}}(\mathbf{A})$ is *coverage-sound* if every valid completion extending $\mathbf{A}$ has each relevant size- s colour class in that family. Equivalently, every predicate used to filter a complete arc universe must be a necessary condition for the corresponding class in a valid completion.

Thus a filtered catalogue is justified in two logically separate steps. The mathematical step proves, conjunct by conjunct, that a valid completion implies the branch predicate. The implementation step checks that the frozen catalogue or prefix ledger is exactly the set defined by that predicate. Independent regeneration is a cross-check of the second step; by itself it is not a substitute for the first.

Lemma 26 (Prefix line capacity). *Let U be the union of r colour classes in an arc-proper six-colouring of G_{11} . Then every grid line L satisfies*

$$|U \cap L| \geq |L| - 2(6 - r).$$

Proof. The remaining $6 - r$ colour classes are arcs, and hence together contain at most $2(6 - r)$ points of L . $\square$

Lemma 27 (Singleton separation). *Two distinct size-21 colour classes in a six-colouring cannot have the same singleton row or the same singleton column.*

Proof. If two size-21 classes had the same singleton row, together they would cover only two points of that row. The other four colours cover at most eight, leaving at most ten covered points, a contradiction. Columns are identical. $\square$

The branch proofs specify the outer prefix predicates and their necessity. For a terminal catalogue, the mathematical reference family is $\mathcal{A}_s(R)$; any filtering must satisfy Definition 25, and complete generation of the filtered family is a separate requirement. The unconditional Q_8/Q_9 and Q_{10} encodings verify the meaning of the corresponding residual problems. Equivalence of these encodings does not by itself establish coverage of an optimized catalogue.

7.3.1 Successive arc decomposition

Large residual classes are selected successively from admissible arc families, reducing the remaining partition after each choice.

For P_4 and P_5 , fixing the three size-21 classes leaves 58 points, with residual profiles

$$20 + 20 + 18 \quad \text{and} \quad 20 + 19 + 19,$$

respectively. Enumerating one admissible size-20 arc therefore leaves only two residual classes.

The same idea is used one level deeper in P_6 . After fixing its two size-21 classes, the residual profile is

$$20 + 20 + 20 + 19.$$

The exact residual search enumerates the admissible size-20 arcs, examines compatible unordered pairs of them, and leaves a 39-point residual of type $20 + 19$.

The Q_8 and Q_9 computations have an analogous shape. After a compatible size-22, size-21 prefix has been fixed, the remaining 78 points have profiles

$$20 + 20 + 20 + 18 \quad \text{or} \quad 20 + 20 + 19 + 19.$$

Two admissible size-20 arcs are selected, reducing the final decision to a two-class residual.

7.3.2 Forced-complement closure

A two-class residual requires no independent search for both classes. If U is to be divided into classes of sizes a and b , with $|U| = a + b$, then after choosing an a -point arc $B \subseteq U$, the second class is forced to be $U \setminus B$. The residual is feasible precisely when this complement is a b -point arc and satisfies any other already proved necessary conditions.

This gives the terminal problems

profile	two-class tail
P_4	$20 + 18$
P_5	$19 + 19$
P_6	$20 + 19$
Q_8	$20 + 18$
Q_9	$19 + 19$.

When the two classes have equal size, exchanging them changes only their labels, so one canonical ordering suffices.

7.3.3 Unlabelled arc packing

When several equal-sized residual classes remain, we instead enumerate an unordered family of pairwise disjoint arcs.

The cleanest example is P_7 . Once its unique size-21 class A is fixed, $|G_{11} \setminus A| = 100$, and the complement must be partitioned into five size-20 arcs:

$$G_{11} \setminus A = B_1 \dot{\cup} \cdots \dot{\cup} B_5, \quad |B_i| = 20.$$

Since the five residual colours have identical roles, only the unordered set $\{B_1, \dots, B_5\}$ matters. Every size-20 colour class has exactly two singleton rows and two singleton columns by Lemma 19, and it meets both main diagonals by Lemma 18. The later P_7 production route therefore uses a reusable global catalogue of size-20 arcs satisfying these universal colour-class conditions, rather than an anchor-specific candidate family. The P_7 computation is an exact five-arc packing problem over this global catalogue.

A closely related, but distinct, formulation is used for Q_{10} . After fixing its size-22 class, the 99-point complement must have profile

$$20 + 20 + 20 + 20 + 19.$$

The four size-20 classes are interchangeable and may be selected as an unordered four-arc packing. Once they have been chosen, the remaining 19 points are forced; a completion exists exactly when this remainder is an arc satisfying the required residual conditions.

The profile eliminations below use these reductions within the three families $\mathcal{R}$, $\mathcal{Q}$, and $\mathcal{P}$.

7.4 Exhaustive elimination of the profile families

We treat $\mathcal{R}$, $\mathcal{Q}$, and $\mathcal{P}$ in turn. The catalogue, symmetry, coverage, and terminal checks are summarised in Section 7.5.

7.4.1 The $\mathcal{R}$ -family

Proposition 28. *No arc-proper six-colouring of G_{11} belongs to $\mathcal{R}$.*

Computational proof. Form the disjointness graph on the 676 admissible size-22 arcs, joining two vertices when the corresponding arcs are disjoint. The exhaustively constructed disjointness graph has 2 138 edges and no triangle. Hence a six-colouring cannot contain three size-22 classes, so every colouring in $\mathcal{R}$ would contain exactly two.

For a disjoint pair $\{A, B\}$ that occurs in a six-colouring, put $R = G_{11} \setminus (A \cup B)$. Necessarily

$$|R \cap \ell| \leq 8 \quad \text{for every grid line } \ell,$$

because the four residual colour classes are arcs and hence together cover at most eight points of any line. This necessary condition is exactly the production residual-capacity filter. Of the 2 138 disjoint unordered pairs, 930 survive it. Joint canonicalisation under the square symmetries and interchange of the two equal-role size-22 classes, that is under $D_4 \times S_2$, leaves 119 canonical pair orbits. Since every valid $\mathcal{R}$ -colouring satisfies the displayed line-capacity condition, this filtering discards no pair that can occur in a valid colouring.

The 119 orbits are the canonical feasible unordered two-22 skeletons. The two size-22 classes give the two zero deficits; the other four deficits are positive and sum to 11, giving 11 nondecreasing residual profiles. Hence the

$$119 \cdot 11 = 1\,309$$

terminal SAT instances cover the whole $\mathcal{R}$ -family. Every instance is unsatisfiable, and every refutation passes independent proof checking. Thus no colouring belongs to $\mathcal{R}$. $\square$

Remark 29. Before using the triangle-free disjointness graph, there are 44 nondecreasing six-part deficit profiles with total deficit 11: 16 have at least three zero deficits, 11 have exactly two, 10 have exactly one, and 7 have none. Triangle-freeness excludes the first 16, leaving the $28 = 11 + 10 + 7$ profiles handled by the $\mathcal{R}$, $\mathcal{Q}$, and $\mathcal{P}$ computations.

7.4.2 The $\mathcal{Q}$ -family

Every profile in $\mathcal{Q}$ contains a unique size-22 class. By Lemma 21, this class may be fixed to one of the 89 representatives of $\mathcal{C}_{22}/D_4$. Its complement contains 99 points, and the ten profiles differ only in the sizes of the five residual colour classes.

Proposition 30 ($\mathcal{Q}$ -family elimination). *None of $Q_1, \dots, Q_{10}$ is realised by an arc-proper six-colouring of G_{11} .*

Computational proof. We first dispose of the seven profiles containing at least two size-21 classes. The frozen reduction package reconstructs the complete 1 019 640-member oriented size-21 catalogue and the 89 representatives of the unique size-22 class.

The profiles Q_1 – Q_4 . Fix the size-22 class and choose three pairwise-disjoint size-21 arcs in increasing mask order. For Q_2, Q_3, Q_4 these are exactly the three size-21 classes; for Q_1 there are four such classes, so choosing any three causes duplication but cannot cause omission. The four fixed classes occupy 85 points and leave 36. Since each of the final two classes has size at most 21, their unordered sizes are exactly

$$21 + 15, \quad 20 + 16, \quad 19 + 17, \quad 18 + 18,$$

which are precisely Q_1, Q_2, Q_3, Q_4 .

At every prefix depth the search uses only pairwise disjointness, Lemma 27, and the necessary line-capacity bound of Lemma 26. The final 36-point residual is an exact two-colour DPLL problem: every residual line has at most two points of either colour, both global class sizes are at most 21, and fixing the least residual point to one residual colour removes only the interchange of the two residual colours.

The six frozen shards cover all 89 size-22 representatives. Their aggregate audit is

stage	count
disjoint first size-21 candidates	1 000 619
after first line-capacity test	412 995
second candidates tested	7 470 942
after second capacity/compatibility test	1 325 039
third candidates tested	10 929
36-point terminal residuals	670.

All 670 residuals are UNSAT. A separately authored verifier reconstructs the same per-anchor prefix counts and the same 670 residual masks, and solves the residuals with an independently organised NAE-triple DPLL rather than the production line-count propagator. It again reports zero satisfiable residuals. Hence $Q_1, \dots, Q_4$ are eliminated without any branch-specific unpublished predicate.

The profiles Q_5 – Q_7 . Fix the size-22 class and choose two size-21 classes. The first candidate is drawn from the complete size-21 catalogue, is disjoint from the anchor, and must satisfy the four-colour residual capacity condition

$$|(A \cup B) \cap L| \geq |L| - 8 \quad (L \text{ a grid line}).$$

Across the 89 anchors there are 1 000 619 disjoint candidates and exactly 412 995 survive this necessary test. The second size-21 class is required to be disjoint, to have different singleton row and column, and to satisfy the corresponding three-colour line-capacity bound. The complete audit tests 7 470 942 second candidates and retains 1 325 039 increasing compatible pairs. Thus exchange of the two equal size-21 colours is removed without deleting a pair.

The three fixed classes use 64 points, leaving 57. The only sorted residual profiles with all three classes of size at most 20 are

$$20 + 20 + 17, \quad 20 + 19 + 18, \quad 19 + 19 + 19,$$

namely Q_5, Q_6, Q_7 . The residual solver is an exact three-colour search with exact quotas and all 628 maximal-line arc constraints. Its propagation rules use only logical upper/lower consequences of the per-line bound two; an exhaustive local audit of 137 200 line-domain states found no false rejection. Equal residual sizes are handled only by first-occurrence value precedence.

For Q_5/Q_6 , the completed reproducibility package records nine pairwise-disjoint shards, normal termination of every shard, no missing, overlapping, timed-out or UNKNOWN work item, and the final audit `FINAL_SHARD_AUDIT PASS workers=9 status=UNSAT`. Q_7 is handled by a separate standalone run over the same audited 1 325 039-prefix universe; its final audit likewise reports complete UNSAT coverage.

The profiles Q_8 and Q_9 . Here there is exactly one size-21 class. Define the retained prefix universe

$$\mathcal{U}_{89} = \{(A, B) : A \in \mathcal{T}_{22}, B \in \mathcal{C}_{21}, A \cap B = \emptyset, |(A \cup B) \cap L| \geq |L| - 8 \ \forall L\}.$$

The formal scan contains $89 \cdot 1\,019\,640 = 90\,747\,960$ anchor–arc pairs. Disjointness leaves 1 000 619, and the displayed line-capacity test leaves exactly 412 995. Lemma 26 proves this filter necessary, so every genuine Q_8 - or Q_9 -colouring has a representative in $\mathcal{U}_{89}$. The 412 995 total and its per-anchor distribution were independently regenerated from the frozen arc catalogues.

There is also an independently audited unconditional base formulation for this branch. For a fixed size-22 anchor, an exact-one ownership encoding selects at most one disjoint size-21 arc and partitions every remaining point among four residual arc colours, each of size at most 20. If s is the selector count, exact ownership gives

$$99 = 21s + \sum_{i=1}^4 |C_i|, \quad s \leq 1, \quad |C_i| \leq 20.$$

Hence $s = 1$, the four residual sizes sum to 78, and their only sorted possibilities are (20, 20, 20, 18) and (20, 20, 19, 19). The formula contains the exact arc constraints and a safe S_4 first-occurrence symmetry break, and contains no dominance strengthening. Consequently it is satisfiable exactly when the fixed anchor extends to Q_8 or Q_9 . This gives an independent semantic reference for the production residual solver.

The production route fixes a prefix in $\mathcal{U}_{89}$, selects the large residual size-20 arcs, and closes the last two colours by forced complement. The completed computation exhausts all 412 995 retained prefixes and finds no feasible residual partition. Since $\mathcal{U}_{89}$ contains a representative of every possible Q_8 - or Q_9 -colouring, this eliminates both profiles.

The profile Q_{10} . Once a size-22 anchor is fixed, the residual has 99 points. A second independently audited unconditional encoding contains no size-21 selector: it assigns each residual point to exactly one of five arc colours, each of size at most 20. Therefore the only possible unordered residual profile is

$$20 + 20 + 20 + 20 + 19.$$

The five residual colours have identical constraints; ordering their first occurrences is an exact S_5 symmetry break. Thus, for each of the 89 anchors, this base formula is satisfiable exactly when the anchor extends to a Q_{10} -colouring. This proves equivalence of the unconditional formulation to the fixed-anchor problem.

For this anchor A , a direct exact formulation selects four pairwise disjoint members of $\mathcal{A}_{20}(G_{11} \setminus A)$ and checks whether the remaining 19 points form an arc. Every Q_{10} colouring supplies such a selection, and every successful selection reconstructs a Q_{10} colouring.

The production computation uses an anchor-relative size-20 catalogue, an unordered four-arc join, and the forced 19-point remainder check. Its catalogue-coverage argument must show that every size-20 class in a possible completion is retained; it need not retain arcs excluded by proved necessary conditions. The completed computation covers all 89 anchors, examines 74 011 954 catalogue masks, and finds no witness. Every join terminates with status

Catalogue coverage, the exact four-arc join and the forced-complement test establish infeasibility of each fixed-anchor problem. The unconditional formulation above provides an independent check of the residual problem’s semantics.

Thus none of the profiles $Q_1, \dots, Q_{10}$ is realisable. $\square$

7.4.3 The $\mathcal{P}$ -family

The profiles in $\mathcal{P}$ contain no size-22 class. The search fixes size-21 classes in the following blocks:

profile block	P_1-P_3	P_4, P_5	P_6	P_7
# fixed size-21 classes	4	3	2	1
residual points	37	58	79	100.

In P_1 , the fifth size-21 class is handled within the residual problem.

Proposition 31 (P-family elimination). *None of $P_1, \dots, P_7$ is realised by an arc-proper six-colouring of G_{11} .*

Computational proof. The profiles P_1-P_3 . Every colouring in these three profiles contains at least four size-21 classes. Choose four of them. They occupy 84 points and leave a 37-point residual. Since no residual class can exceed 21, its two unordered class sizes are exactly

$$21 + 16, \quad 20 + 17, \quad 19 + 18,$$

which are precisely P_1, P_2, P_3 . Conversely every colouring of one of these profiles supplies such a four-arc prefix.

The first size-21 arc is taken from the 127 491 individual D_4 -representatives and the remaining three mask IDs are increasing. The global-minimum argument under the common D_4 action shows that every unordered four-arc packing has such a representative. Pairwise disjointness, Lemma 27, and the prefix capacity bound of Lemma 26 are the only mathematical prefix restrictions. After four arcs, every residual line has at most four points.

The residual is solved exactly as a two-colour Boolean problem, with at most two points of either colour on every line and global size at most 21. Fixing the least residual point to one colour removes only the residual colour swap. Six shards partition the 127 491 first-arc representatives by index modulo six. The complete aggregate is

stage	count
first arcs	127 491
depth-2 survivors	514 209 820
depth-3 survivors	1 505 452 322
depth-4 / terminal residuals	708 638.

All six shards exit normally with zero satisfiable residuals. A separately authored verifier reconstructs the arc list and maximal lines by a different representation, re-enumerates every prefix, streams the same residual masks, and solves each by NAE-triple DPLL rather than the production line-count solver. This independently checks both prefix coverage and the terminal decisions for P_1, P_2, P_3 .

The profiles P_4 and P_5 . The three size-21 classes are interchangeable. Before profile-transfer pruning, the raw geometric outer objects are fixed-first occurrences of unordered triples $T = \{A, B, C\} \subset \mathcal{C}_{21}$ whose members are pairwise disjoint, have pairwise distinct singleton rows and singleton columns, and satisfy the $r = 3$ instance of Lemma 26. These are necessary conditions for the three size-21 classes of either profile.

The nonextendibility reduction is also explicit. In P_4 , a one-point extension of a size-21 recipient from a donor of size 21, 20, 18 would produce respectively Q_8, Q_6, Q_5 . In P_5 , donors of size 21, 20, 19 would produce respectively Q_9, Q_7, Q_6 . Since those target profiles have already been eliminated, Lemma 23 shows that every size-21 class occurring in a P_4 or P_5 colouring is globally nonextendible. The frozen full-count audit gives the complete chain

$$1\,699\,467\,737 \xrightarrow{\text{certified nonextendibility/dependency filter}} 1\,633\,664\,146 \xrightarrow{/(D_4 \times S_3)} 544\,481\,464.$$

The first number counts raw compatible, duplicate-retaining arc-triple occurrences; the second is the complete dependency-reduced occurrence count; the third is the number of canonical arc-triple records after exact joint $D_4 \times S_3$ canonical augmentation. None of these numbers counts distinct residual masks.

For completeness, the frozen exactly-one-orbit argument can be stated directly. Order the 121-bit arc masks as unsigned integers and, for an unordered triple $T = \{A, B, C\}$, define

$$K_3(T) = \min_{g \in D_4} \text{sort}(gA, gB, gC),$$

where the sorted triple is compared lexicographically. A fixed-first occurrence (F, A, B) is retained exactly when $F < A$, $F < B$, and $\text{sort}(F, A, B) = K_3(T)$. Write $K_3(T) = (k_0, k_1, k_2)$ with $k_0 < k_1 < k_2$. The first member k_0 is itself D_4 -canonical: otherwise a symmetry sending k_0 below itself would send the sorted triple below the defining minimum $K_3(T)$. Hence the audited fixed-first list contains k_0 , and its unordered pair loop contains $\{k_1, k_2\}$. All geometric and nonextendibility conditions above are D_4 -invariant, so this occurrence survives. Conversely, every retained occurrence has sorted tuple equal to the unique $K_3(T)$; its distinguished member must be the strict minimum k_0 , and the unordered pair is emitted once. Thus exactly one representative of every admissible $D_4 \times S_3$ orbit is retained. Square symmetries preserve arc validity, disjointness and the residual geometry, while S_3 merely relabels the three equal size-21 colours, so feasibility is constant on each orbit.

Each retained triple leaves 58 points, with residual profile

$$P_4 : \quad 20 + 20 + 18, \quad P_5 : \quad 20 + 19 + 19.$$

The residual solver chooses one required large arc and closes the final two colours by forced complement. The completed terminal campaign exhausts all 544 481 464 canonical records and returns UNSAT for both profiles. The raw-universe, transfer and orbit arguments above establish coverage of every possible colouring with either profile.

The profile P_6 . The two size-21 classes are interchangeable, so their outer universe is a joint $D_4 \times S_2$ orbit problem, not two independently reduced arc orbits. A completed global pair-enumeration audit reports the following stage counts:

stage	ordered oriented	unordered oriented	$D_4 \times S_2$ canonical
all distinct pairs	1 039 664 709 960	519 832 354 980	64 979 365 359
disjoint	14 835 875 072	7 417 937 536	927 247 231
singleton-row/column compatible	11 294 847 960	5 647 423 980	705 931 816
structurally compatible	4 362 894 528	2 181 447 264	272 681 893
after certified Q_8/Q_9 transfers	4 268 334 336	2 134 167 168	266 771 874.

The global enumeration uses five disjoint production shards covering all 127 491 first-arc representatives. An independent Burnside count over all 1 019 640 oriented arcs agrees at every D_4 -invariant stage, including the contribution of nontrivial stabilisers.

The structural predicate is as follows. Let $s_r(A)$ and $s_c(A)$ be the singleton row and column of $A \in \mathcal{C}_{21}$. The pre-transfer P6 pair universe is

$$\begin{aligned} \mathcal{U}_6^{\text{str}} = \{ \{A, B\} \in \mathcal{C}_{21} : & A \cap B = \emptyset, \quad s_r(A) \neq s_r(B), \\ & s_c(A) \neq s_c(B), \\ & |(A \cup B) \cap L| \geq |L| - 8 \text{ for every grid line } L \}. \end{aligned}$$

Every genuine P6 pair belongs to $\mathcal{U}_6^{\text{str}}$: membership in $\mathcal{C}_{21}$ follows from long-line rigidity, disjointness from being colour classes, singleton separation from Lemma 27, and the final inequality from Lemma 26 with $r = 2$.

The exact joint orbit key is

$$K_2(A, B) = \min_{g \in D_4} \text{sort}(gA, gB),$$

with the same symmetry applied to both arcs; sorting quotients the exchange of the two equal size-21 colours. Thus stabilisers, size-4 and size-8 arc orbits, same-individual-orbit pairs, and pair exchange are all handled by one group action. The 272 681 893 structural count in the table is the number of K_2 -orbits of $\mathcal{U}_6^{\text{str}}$. The independent Burnside enumeration provides a separate exact check of this orbit count; the mathematical coverage claim itself follows from the displayed predicate and joint orbit key.

The final transfer stage is narrower and can be stated mathematically. For either fixed size-21 recipient, moving an admissible point from a residual size-20 donor produces profile Q_9 , while moving an admissible point from the residual size-19 donor produces Q_8 :

$$Q_9 : (22, 21, 20, 20, 19, 19), \quad Q_8 : (22, 21, 20, 20, 20, 18).$$

After Q_8, Q_9 have been eliminated, Lemma 23 therefore excludes these donor-to-recipient moves. For an arc A , define

$$\text{Ext}(A) = \{v \in G_{11} \setminus A : A \cup \{v\} \text{ is an arc}\}.$$

For a structural pair $\{A, B\}$, put $R = G_{11} \setminus (A \cup B)$. A necessary condition for this pair to occur in a P_6 colouring is

$$R \cap (\text{Ext}(A) \cup \text{Ext}(B)) = \emptyset. \quad (14)$$

Indeed, in any completion every point of R belongs to a donor of size 20 or 19, so an extension of either recipient would produce Q_9 or Q_8 . Points in the other size-21 class are outside this test. Thus the condition does not impose global nonextendibility or use Q_{10} pruning.

Every retained pair leaves a 79-point residual of type $20 + 20 + 20 + 19$. The exact residual computation selects the size-20 classes, quotients only the interchange of the three equal residual colours, and ends with a forced $20 + 19$ decision. The completed computation exhausts all 266 771 874 retained canonical pairs, and none admits a valid completion.

The profile P_7 . There is a unique size-21 class, which may be fixed to one of the 127 491 arcs in $\mathcal{T}_{21}$. Its complement contains 100 points and must be partitioned into five interchangeable size-20 arcs.

For clarity, define the global size-20 colour-class universe

$$\begin{aligned} \mathcal{C}_{20}^{\text{glob}} = \{ & B \subseteq G_{11} : |B| = 20, \quad B \text{ is an arc,} \\ & 1 \leq |B \cap R_i| \leq 2 \text{ for every row } R_i, \\ & 1 \leq |B \cap K_j| \leq 2 \text{ for every column } K_j, \\ & B \text{ meets both main diagonals} \}. \end{aligned}$$

Because B has twenty points, the row and column conditions are equivalent to saying that it has exactly two singleton rows and exactly two singleton columns. Lemmas 18 and 19 show that every size-20 colour class in a hypothetical six-colouring belongs to $\mathcal{C}_{20}^{\text{glob}}$.

The production computation materialised a global store of 158 957 570 sorted unique oriented members of $\mathcal{C}_{20}^{\text{glob}}$. The catalogue theorem proves that its 55 generation shards are pairwise disjoint and exhaustive for this explicitly defined universe. A second implementation

reproduces the entire store record-for-record. The catalogue therefore has the explicit global scope displayed above.

For every anchor the terminal question is exactly the unordered packing

$$G_{11} \setminus A = B_1 \dot{\cup} \dots \dot{\cup} B_5, \quad |B_i| = 20.$$

The completed packing computation covers all 127 491 anchors and finds no feasible five-arc packing. The displayed definition, long-line rigidity and the deficit–singleton lemma ensure that $\mathcal{C}_{20}^{\text{glob}}$ contains every possible residual size-20 colour class, and the 55-shard catalogue exhausts this universe.

Hence none of the profiles $P_1, \dots, P_7$ is realisable. $\square$

7.5 Computational certification

The branch proofs establish that every hypothetical six-colouring reaches one of the finite search domains in Table 4. They give the necessary filtering predicates, the joint symmetry reductions and the residual partition problems. The completed computational ledgers account for every work item in these domains, with no valid completion.

Case	Search domain and count	Terminal decision
$\mathcal{R}$	2 138 disjoint unordered pairs $\rightarrow$ 930 capacity-feasible pairs $\rightarrow$ 119 pair orbits; 119 $\cdot$ 11 = 1 309 instances	SAT instances for the 11 residual profiles: all UNSAT, with checked refutations
Q_1 – Q_4	670 residuals of 36 points from 89 anchors	Exact two-colour search: all UNSAT
Q_5, Q_6	1 325 039 prefixes of sizes 22, 21, 21	Exact three-colour search for 20 + 20 + 17 and 20 + 19 + 18: all UNSAT
Q_7	The same 1 325 039 prefixes	Separate exact search for 19 + 19 + 19: all UNSAT
Q_8, Q_9	412 995 retained prefixes of sizes 22, 21	Residual arc selection and forced complement: no feasible partition
Q_{10}	89 anchors, each with a 99-point residual	Four size-20 arcs and a forced size-19 complement: no feasible partition
P_1 – P_3	708 638 residuals of 37 points from 127 491 first-arc representatives	Exact two-colour search: all UNSAT
P_4, P_5	544 481 464 canonical triples of size-21 arcs	Residual arc selection and forced complement for both profiles: all UNSAT
P_6	266 771 874 canonical pairs after the Q_8/Q_9 transfer exclusions	Residual arc selection and forced 20 + 19 completion: no feasible partition
P_7	127 491 anchors, each with a 100-point residual	Unordered packing of five size-20 arcs from $\mathcal{C}_{20}^{\text{glob}}$: no feasible partition

Table 4: Completed computations excluding the six-colour profiles of G_{11} . Counts refer to the stated search domains; the branch proofs describe their construction and the candidate catalogues.

Validation is specific to each computation. The $\mathcal{R}$ refutations were independently proof-checked. Separate implementations re-enumerated all prefixes and re-solved every residual for Q_1 – Q_4 and P_1 – P_3 . The Q_5/Q_6 runs have a complete nine-shard terminal audit, and Q_7 has a separate terminal audit over the same audited prefix universe. Independent enumeration checks the Q_8/Q_9 prefixes, Burnside enumeration checks the P_6 orbit counts, and a second implementation reproduces the global P_7 catalogue record-for-record. The unconditional Q_8/Q_9 and Q_{10} encodings independently check that the residual models express the intended partitions; the production exclusions also use catalogue coverage and exhaustive residual

searches. The completed ledgers and branch-specific terminal audits cover the remaining searches in the table. Only exact terminal decisions establish infeasibility; timeouts and UNKNOWN statuses do not. A *proof object* is a formal refutation or an independently checkable exhaustive certificate; an execution log alone records terminal status.

7.6 Completion of the theorem

Theorem 32. *The arc chromatic number of the 11×11 grid is*

$$\chi_{\mathcal{A}}(G_{11}) = 7.$$

Proof. By Propositions 28, 30, and 31, no arc-proper six-colouring belongs to $\mathcal{R}$, and none has any of the profiles $Q_1, \dots, Q_{10}$ or $P_1, \dots, P_7$. By Theorem 20, these cases exhaust all possible arc-proper six-colourings of G_{11} . Hence no such colouring exists and $\chi_{\mathcal{A}}(G_{11}) \geq 7$. The G_{12} colouring gives $\chi_{\mathcal{A}}(G_{11}) \leq 7$, so $\chi_{\mathcal{A}}(G_{11}) = 7$. $\square$

8 Explicit upper-bound certificates

For the upper bounds used in Theorems 3 and 5, the colour of (x, y) is the entry in column x and displayed row y of the corresponding matrix below. Thus the displayed rows are part of the certificate, not merely an illustration. Relabelling colours or applying a square symmetry would give an equivalent certificate, but the matrices below are the frozen versions used for the stated digest and verification data.

G_9 .

1	2	2	5	1	4	4	3	5
4	5	1	5	3	2	2	1	3
3	5	4	3	4	5	1	2	1
3	1	2	1	5	4	3	4	5
5	4	3	3	2	1	1	4	2
5	1	1	2	5	3	4	3	2
2	4	3	4	2	1	5	5	1
1	3	5	4	1	5	3	2	4
4	2	5	2	4	3	5	1	3

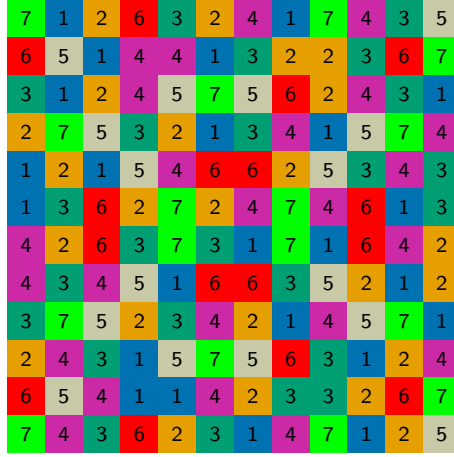
Its colour-class sizes, in label order $1, \dots, 5$, are $(17, 15, 16, 16, 17)$. The independent normalized-line verifier gives the same set of collinear triples.

G_{10} .

1	2	5	2	1	3	4	5	4	3
5	1	4	1	2	4	3	2	3	5
3	2	3	5	2	4	5	1	4	1
4	6	4	3	1	3	1	2	6	2
3	4	3	4	6	6	2	1	2	1
2	1	2	1	6	6	3	4	3	4
1	5	1	2	4	2	4	3	5	3
2	3	2	5	3	1	5	4	1	4
6	4	1	4	3	1	2	3	2	6
4	3	6	3	4	2	1	6	1	2

Its class sizes are $(20, 20, 20, 20, 10, 10)$.

G_{12} .



Its class sizes are (24, 24, 24, 24, 16, 16, 16).

For reproducibility, the canonical text representation of each certificate uses one ASCII space between entries and a line feed after every row, including the last. The SHA-256 digests of those exact text files are

certificate	SHA-256
G_9	43558895f3ed58cd7e8b7410742c28089526be9ff69fb22cdedf72a7866bf409
G_{10}	96efaf6aca921323fbdeeeb77d0dd54407b6ecb2bf786a2f457466d75912876a
G_{12}	a379c6caf103cb60fce8c8baab051399418920cd013bf6df19f5dc5988a094a3

The reference outputs are:

Grid	Unordered triples	Collinear	Monochromatic collinear
G_9	85 320	2 712	0
G_{10}	161 700	4 448	0
G_{12}	487 344	10 332	0

9 Computational supplement

The supplement contains the frozen inputs, source code, arc catalogues, exhaustive work-item ledgers, terminal outputs, file hashes and validation material. Its claim-to-artifact ledger identifies the files supporting each computational claim.

The upper-bound witnesses for G_9 , G_{10} , G_{12} , their SHA-256 digests and reference verification counts are given in Section 8.

The branch predicates, orbit arguments and completed G_{11} searches are described in Sections 7.4 and 7.5. Table 4 gives their work-item counts and validation methods. Source and binary hashes, shard partitions and detailed execution records are retained in the supplement.

The overall evidence map is:

Claim	Primary evidence	Validation / cross-check
$\chi_{\mathcal{A}}(G_9) = 5$	Explicit matrix	Determinant, normalized-line and separate C++ triple verifiers
$C(5) = 62$, $C(7) = 1\,726$	Representative catalogues and count manifests	Validity, colour/square canonicalisation and orbit/count audits
$C(6)$	Per-profile meet-in-the-middle and fixed-point ledger	Arithmetic audit of all 33 contributions; not a second enumeration
$C(8) = 2$	Full-arc catalogue and four unordered exact covers	Arc/orbit checks and canonicalisation of the partitions
$\chi_{\mathcal{A}}(G_{10}) = 6$, $\chi_{\mathcal{A}}(G_{12}) = 7$	Full-arc catalogues, 26-case corner obstruction, exact-cover exclusion and witness matrices	Input/orbit audits, checked G_{10} refutations and matrix verification
$\chi_{\mathcal{A}}(G_{11}) = 7$ $C(9) = 743$	Profile reductions and Table 4 Exhaustive arc-generation enumeration and representative catalogue	Branch-specific checks in Section 7.5 Independent exhaustive enumeration, validity and canonicalisation
Packing bounds	Full-arc inputs and exact packing exclusions for $n = 14, 16, 18, 20$	Input/coverage validation and replay of terminal outcomes

Code and data availability. The completed computational packages will be made publicly available in the [GitHub repository for this paper](#). Jason Dong is assembling the repository release.

Discovery disclosure. One G_9 colouring was first found by an AI-assisted randomized search. No theorem depends on that search trajectory: the stated results use only the explicit witness, exhaustive enumeration, and subsequent verification. The authors take responsibility for all claims in the manuscript.

10 Conclusion

We establish $\chi_{\mathcal{A}}(G_{11}) = \chi_{\mathcal{A}}(G_{12}) = 7$ and provide constructions and proofs for the previously announced values $\chi_{\mathcal{A}}(G_9) = 5$ and $\chi_{\mathcal{A}}(G_{10}) = 6$. The full-arc packing computations improve the row bound for $n = 14, 16, 18, 20$; the adjacent odd-grid bounds already follow from the row bound. The small-grid classification gives $C(6) = 2\,949\,015\,889$, $C(8) = 2$, and $C(9) = 743$. For infinitely many n we have $\chi_{\mathcal{A}}(G_n) \leq n - 8$; whether $n - \chi_{\mathcal{A}}(G_n)$ is unbounded remains open.

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